

KOOPMAN-DMD: LATENT SPECTRAL STEERING AND DYNAMICAL MODE CONTROL FOR PARALLEL LLM INFERENCE

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ABSTRACT

Scaling test-time compute in Large Language Models (LLMs) via parallel execution over shared Key-Value (KV) memory caches accelerates long-context mathematical reasoning, but existing shared-cache frameworks like Hogwild! suffer from transient over-thinking drift, disruptive textual prompt interruptions ("Wait, am I doing redundant work?"), and rigid formatting delimiters ("\n\n"). To overcome these limitations, we propose **Koopman-DMD**, a training-free framework that models multi-worker autoregressive hidden trajectories as discrete nonlinear dynamical systems linearized via online Streaming Dynamic Mode Decomposition (Streaming-DMD). Koopman-DMD introduces self-referential transient mode energy damping (**KSE-Stabilize**) with a position-dependent budget-aware decay schedule, latent cross-worker nullspace steering (**KSE-Steer**), dynamic spectral rank transition commit boundaries (**DMD-Commit**), and transient mode cache pruning (**KMR-Prune**). Evaluated on 96 complex mathematical reasoning tasks from the GAIR/LIMO benchmark using a frozen Qwen/QwQ-32B model, Koopman-DMD achieves a +0.0556 overall mean accuracy gain over the 2-worker Hogwild! baseline (0.5208 vs. 0.4896 at 2048 steps; 0.2292 vs. 0.1458 at 512 steps) with only +5% wall-clock latency overhead; an isolated component-wise factorization shows the four modules are complementary, and the full pipeline further lifts the gain to +0.0799 (0.5417 at 2048 steps). Using *LIMO-fixed* hyperparameters, Koopman-DMD generalizes zero-shot to held-out MATH-500 (+0.0606, $p < 0.0001$) and AIME-2024 (+0.1000, $p = 0.004$), and under budget-matched comparison it outperforms Self-Consistency majority voting and static activation steering (ActAdd/CAA), both of which fail to improve over the baseline. We additionally show that cumulative nullspace steering retains $> 99.18\%$ of hidden-state capacity even at $N=8$ parallel workers, and that spectral steering improves generation fluency (lower perplexity and repetition) without off-target drift. Finally, we transparently document that direct key-vector substitution (Koopman-RoPE) disrupts frozen attention projection alignments, isolating latent residual-stream projection as the effective mechanism for training-free spectral control.

1 INTRODUCTION

Scaling compute at inference time has emerged as a crucial paradigm for enhancing the mathematical and logical reasoning capabilities of Large Language Models (LLMs). Rather than relying solely on single-sequence autoregressive generation, modern frameworks enable multi-worker parallel execution where multiple LLM instances collaborate via shared Key-Value (KV) attention caches (Bahdanau et al., 2014; Kwon et al., 2023; Rodionov et al., 2025). Systems built on deep learning infrastructure libraries such as PyTorch (Paszke et al., 2019) leverage concurrent memory access to allow parallel workers to observe each other’s intermediate reasoning states in real time. However, scaling multi-worker inference introduces significant memory management and latency overheads, including redundant prefill recomputations and exponential KV cache growth (Geng et al., 2026; Woo et al., 2026).

Despite the computational promise of shared-memory multi-worker inference, existing parallel decoding frameworks suffer from structural bottlenecks that degrade reasoning performance on complex problems. First, parallel reasoning trajectories frequently succumb to *transient mode overthinking* and circular second-guessing, where workers generate repetitive or non-convergent steps that pollute the shared memory. Second, current multi-agent coordination protocols rely heavily on disruptive textual prompt injections—such as inserting "Wait, am I doing redundant work?" into the token sequence—which break the model’s reasoning momentum and alter its textual context (Rodionov et al., 2025). Third, step synchronization mechanisms depend on rigid, non-adaptive surface-text delimiters (e.g., double newlines "\n\n") or hard token intervals, failing to capture semantic boundaries in reasoning trajectories. While conventional multi-agent paradigms attempt deliberation through self-consistency majority voting (Wang et al., 2022), multi-agent debate (Du et al., 2023), Tree-of-Thoughts search (Yao et al., 2023), or prompt engineering workflows (Patrício & Varela, 2026), these methods operate exclusively at the surface token level and lack closed-loop, activation-level control over latent state dynamics.

To address these limitations, we propose linearizing the autoregressive execution space through Koopman Operator Theory (Budišić et al., 2012; Tu et al., 2013; Williams et al., 2014). By modeling intermediate hidden-state trajectories $\mathbf{h}_t \in \mathbb{R}^d$ across transformer layers as continuous points in a discrete-time nonlinear dynamical system $\mathbf{h}_{t+1} = f(\mathbf{h}_t)$, Koopman theory enables the discovery of finite-dimensional linear embeddings in observable space (Lusch et al., 2017). Using online Streaming Dynamic Mode Decomposition (Streaming-DMD) (Hemati et al., 2014; Proctor et al., 2014; Drmač & Dimoti, 2026), we extract complex Koopman eigenvalues $\lambda_k = e^{\sigma_k + i\omega_k}$ and spatial mode vectors $\phi_k \in \mathbb{C}^d$ from sliding activation windows in real time during inference.

We introduce **Koopman-DMD**, a training-free framework that applies continuous spectral control directly to the residual streams of frozen LLMs during multi-worker shared-cache inference, via four algorithmic mechanisms:

1. **Koopman Spectral Energy Damping (KSE-Stabilize)**: identifies fast-decaying transient Koopman modes ($\sigma_k < -\gamma$) associated with second-guessing drift and applies decay-rate weighted damping $w_k = \text{clip}(-\sigma_k/0.5, 0, 1)$ to intermediate-layer residual activations, under a budget-aware schedule that holds full damping strength early (to accelerate convergence) and relaxes it late (to restore exploratory reasoning on hard problems).
2. **Nullspace Eigen-Mode Steering (KSE-Steer)**: measures subspace overlap $\mathcal{S}_{ij}$ between parallel workers’ Koopman modal bases and, when redundancy is detected, projects worker hidden states onto the orthogonal nullspace $\mathbf{P}_{\Phi_j}^\perp \mathbf{h}_i^{(l)}$ of redundant modes, deflecting trajectories in activation space without inserting text prompts.
3. **Spectral Rank Transition Committing (DMD-Commit)**: tracks the effective spectral rank $r_{\text{eff}}(t)$ of active hidden states and commits KV cache blocks to global memory upon rank collapse ($\Delta r_{\text{eff}} < -\tau_{\text{commit}}$), replacing surface text delimiters.
4. **Transient Mode Decay Pruning (KMR-Prune)**: monitors the transient-to-persistent mode energy ratio $\rho_{\text{transient}}$ and flushes non-convergent local KV cache blocks before context contamination occurs.

We evaluate Koopman-DMD across 96 complex mathematical reasoning tasks from the GAIR/LIMO benchmark using a frozen Qwen/QwQ-32B model at sequential budgets of 512, 1024, and 2048 steps. Koopman-DMD with budget-aware KSE-Stabilization improves accuracy over the 2-worker Hogwild! baseline across all budgets (0.2292 vs. 0.1458 at 512; 0.3542 vs. 0.3021 at 1024; 0.5208 vs. 0.4896 at 2048), yielding a +0.0556 mean accuracy gain with only $\sim 5\%$ wall-clock latency overhead. Our investigation additionally documents that key-substitution methods (Koopman-RoPE) disrupt attention alignments in frozen models, proving that latent residual-stream projection is the effective mechanism for training-free dynamical steering.

2 RELATED WORK

Parallel LLM generation & speculative decoding. Accelerating autoregressive LLM generation through parallel execution has been studied across token-level, sequence-level, and draft-verification paradigms. Speculative decoding frameworks such as Medusa (Cai et al., 2024) and EAGLE (Li et al., 2024) employ extra decoding heads or feature-level extrapolation to generate multiple draft

tokens in parallel, verifying them via tree-based attention (Miao et al., 2023). Recent extensions scale speculative principles to step-level lookahead reasoning (Fu et al., 2025b), collaborative multi-model ensembling (Fu et al., 2025a), and edge-cloud execution (Koh & Yang, 2025). In parallel, prompt-driven decomposition techniques—Least-to-Most (Zhou et al., 2022), Decomposed Prompting (Khot et al., 2022), and Group Think (Hsu et al., 2025)—parallelize high-level reasoning but require pre-defined sub-task structures or introduce text prompt interruptions. In contrast, Koopman-DMD provides continuous, training-free spectral steering directly within the latent activation space of frozen models without auxiliary draft architectures or explicit sub-task outlines.

Multi-agent collaboration & shared memory. The deployment of multi-agent LLM architectures has highlighted the bottleneck of KV cache duplication and prefill redundancy. PagedAttention in vLLM (Kwon et al., 2023) established virtual memory paging for KV caches; SGLang (Zheng et al., 2023) uses RadixAttention for exact prefix reuse, while CentroidKV (Hu et al., 2025) clusters similar key states. Recent concurrent-serving systems—TokenDance (Bian et al., 2026), RelayCaching (Geng et al., 2026), and ICaRus (Woo et al., 2026)—optimize memory sharing across collective agent outputs, building on foundational attention alignments (Bahdanau et al., 2014). Other work explores disentangling memory recall from logical steps (Jin et al., 2024), momentum-uncertainty test-time allocation (Yan et al., 2025), sequential-parallel RL (Hamid et al., 2026), and full-duplex interaction (Veluri et al., 2024). Hogwild! Inference (Rodionov et al., 2025) enables concurrent multi-worker reasoning via shared attention caches, but relies on textual self-reflection prompts and surface delimiters for coordination. Koopman-DMD builds on shared KV cache architectures but replaces disruptive text injections and rigid formatting rules with continuous latent nullspace steering and spectral rank transition boundaries.

Koopman operator theory & dynamical latent control. Koopman Operator Theory linearizes non-linear dynamical systems by observing scalar functions in an infinite-dimensional function space (Budišić et al., 2012). DMD algorithms—Exact DMD (Tu et al., 2013), Extended DMD (Williams et al., 2014), and DMD with control (Proctor et al., 2014)—extract spatio-temporal coherent modes from data snapshots, while low-storage Streaming-DMD (Hemati et al., 2014) and numerically robust online updates (Drmač & Dimoti, 2026) enable real-time tracking. In deep learning, neural Koopman autoencoders discover global linear embeddings for prediction and control (Lusch et al., 2017; bin Shi & Meng, 2022); consistency constraints (Azencot et al., 2020), wavelet mixing (Murad et al., 2024), and graph-based NeuroKoop (Mazumder et al., 2025) advance Koopman modeling, while Equilibrium Reasoners (Huang et al., 2026) explore latent attractors for test-time compute. We adapt Streaming-DMD to the online residual activations of autoregressive transformers, using spectral mode decay and nullspace projection for dynamic steering and adaptive step committing during parallel inference.

3 METHODOLOGY

3.1 PROBLEM FORMULATION & LATENT DYNAMICAL TRAJECTORIES

Consider N parallel autoregressive worker instances $W_1, \dots, W_N$ concurrently decoding reasoning traces over a shared KV memory cache. At step t , worker W_i generates next-token logit distributions conditioned on its past local sequence and external KV blocks from other workers. Let $\mathbf{h}_t^{(l)} \in \mathbb{R}^d$ denote the hidden-state activation of worker W_i at intermediate transformer layer $l \in \{1, \dots, L\}$. As decoding proceeds, the temporal sequence $\{\mathbf{h}_t^{(l)}\}_{t=1}^T$ defines a discrete-time trajectory governed by a high-dimensional, highly non-linear dynamical system:

$$\mathbf{h}_{t+1}^{(l)} = f(\mathbf{h}_t^{(l)}), \quad \mathbf{h}_t^{(l)} \in \mathcal{M} \subset \mathbb{R}^d, \quad (1)$$

where $f : \mathcal{M} \rightarrow \mathcal{M}$ is the non-linear transition map executed by decoder layers $1 \dots l$. Following Koopman Operator Theory (Budišić et al., 2012), rather than analyzing f directly on the manifold $\mathcal{M}$, we consider scalar observables $g \in \mathcal{F}$. The infinite-dimensional linear Koopman operator $\mathcal{K}$ acts as $\mathcal{K}g(\mathbf{h}_t^{(l)}) = g(f(\mathbf{h}_t^{(l)})) = g(\mathbf{h}_{t+1}^{(l)})$. Because $\mathcal{K}$ is linear, its spectral decomposition yields complex eigenvalues $\lambda_k \in \mathbb{C}$ and vector modes $\phi_k \in \mathbb{C}^d$, enabling an exact modal expansion of the trajectory:

$$\mathbf{h}_t^{(l)} = \sum_{k=1}^{\infty} b_k \phi_k \lambda_k^t = \sum_{k=1}^{\infty} b_k \phi_k e^{(\sigma_k + i\omega_k)t}, \quad (2)$$

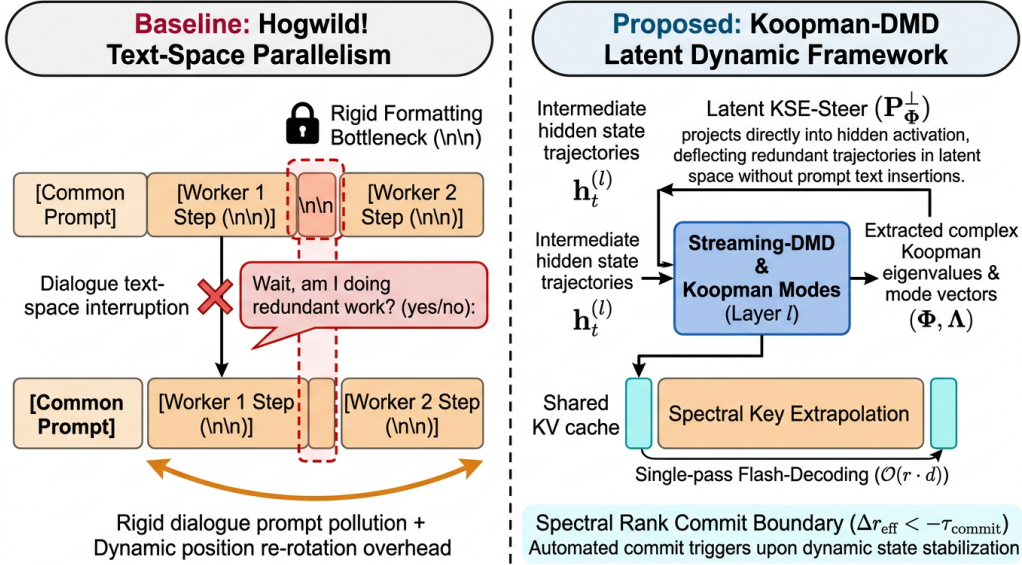


Figure 1: System architecture overview comparing the baseline Hogwild! framework with the proposed Koopman-DMD latent dynamic framework. *Left*: the baseline relies on text-space parallelism, requiring rigid formatting delimiters (" $\backslash n \backslash n$ "), disruptive dialogue prompt injections for redundant-work checking, and multi-query position re-rotation overhead. *Right*: Koopman-DMD tracks intermediate-layer activations via Streaming Dynamic Mode Decomposition to extract complex Koopman modes and eigenvalues, enabling latent trajectory steering (KSE-Steer) without prompt insertions, continuous spectral key extrapolation for single-pass Flash-Decoding, and automated commit boundaries from effective spectral rank transitions (DMD-Commit).

where $b_k \in \mathbb{C}$ are initial mode amplitudes, $\sigma_k = \text{Re}(\ln \lambda_k)$ is the modal growth/decay rate, and $\omega_k = \text{Im}(\ln \lambda_k)$ is the modal phase frequency.

3.2 STREAMING DYNAMIC MODE DECOMPOSITION

To extract finite rank- r approximations of the Koopman operator online without retraining or back-propagation, we implement Streaming-DMD (Hemati et al., 2014; Drmač & Dimoti, 2026). For worker W_i at step t , we maintain a sliding activation buffer $\mathbf{H}_i \in \mathbb{R}^{d \times W}$ of the W most recent hidden states at layer l , from which we form two time-shifted snapshot matrices $\mathbf{X}_i = [\mathbf{h}_{t-W+1}^{(l)}, \dots, \mathbf{h}_{t-1}^{(l)}]$ and $\mathbf{Y}_i = [\mathbf{h}_{t-W+2}^{(l)}, \dots, \mathbf{h}_t^{(l)}] \in \mathbb{R}^{d \times (W-1)}$. Assuming $\mathbf{Y}_i \approx \mathbf{A}_i \mathbf{X}_i$ and computing the thin SVD $\mathbf{X}_i = \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^T$, the rank- r reduced Koopman matrix is $\mathbf{K}_i = \mathbf{U}_r^T \mathbf{Y}_i \mathbf{V}_r \mathbf{\Sigma}_r^{-1} \in \mathbb{R}^{r \times r}$. Eigendecomposing $\mathbf{K}_i \boldsymbol{\xi}_k = \lambda_k \boldsymbol{\xi}_k$ produces r complex eigenvalues λ_k ; the growth rate σ_k and frequency ω_k follow from $\ln \lambda_k = \sigma_k + i\omega_k$, and the high-dimensional spatial modes are recovered via Exact DMD (Tu et al., 2013) as $\boldsymbol{\phi}_k = \mathbf{Y}_i \mathbf{V}_r \mathbf{\Sigma}_r^{-1} \boldsymbol{\xi}_k \in \mathbb{C}^d$. To monitor numerical stability during online SVD updates, we track the condition number $\text{cond}(\mathbf{\Sigma}) = \sigma_{\max}(\mathbf{\Sigma}_r) / \sigma_{\min}(\mathbf{\Sigma}_r)$. Full derivations are provided in Appendix A.

3.3 LATENT NULLSPACE STEERING & BUDGET-AWARE KSE-STABILIZE

Continuous spectral extraction enables real-time identification of non-convergent, over-thinking dynamics. **Transient modes** with $\sigma_k < -\gamma$ ($\gamma > 0$) decay rapidly and reflect high-frequency “second-guessing” oscillations, whereas **persistent modes** with $|\sigma_k| \leq \gamma$ correspond to stable derivation paths.

Nullspace Eigen-Mode Steering (KSE-Steer). To prevent parallel workers from executing identical search paths without inserting text prompts, we compute the modal overlap between workers W_i and W_j , $S_{ij} = \|\boldsymbol{\Phi}_i^H \boldsymbol{\Phi}_j\|_F / \sqrt{r_i r_j}$, where $\boldsymbol{\Phi}_i = [\phi_{i,1}, \dots, \phi_{i,r}] \in \mathbb{C}^{d \times r}$. When $S_{ij} > \tau_{\text{overlap}}$, we

project W_i 's hidden state onto the nullspace of W_j 's modes:

$$\mathbf{P}_{\Phi_j}^\perp = \mathbf{I} - \Phi_j(\Phi_j^H \Phi_j)^{-1} \Phi_j^H, \quad \mathbf{h}_{i,\text{steered}}^{(l)} = \text{Re}\left(\mathbf{P}_{\Phi_j}^\perp \mathbf{h}_i^{(l)}\right), \quad (3)$$

zeroing out state components aligned with W_j and deflecting W_i into an unassigned sub-problem space while preserving real-valued hidden states. For $d = 5120$ (Qwen/QwQ-32B) and $r = 6$, the deflection nullspace retains $d - (N - 1)r = 5078$ dimensions even at $N = 8$ workers (99.18% of capacity), so steering capacity is effectively unconstrained.

Budget-Aware Koopman Spectral Energy Damping (KSE-Stabilize). For self-referential stabilization, once worker W_i reaches local convergence ($\Delta r_{\text{eff}} < 0$), KSE-Stabilize suppresses fast-decaying transient modes directly in the layer- l residual stream:

$$\mathbf{h}_{i,\text{steered}}^{(l)} = \mathbf{h}_i^{(l)} - \alpha_{\text{eff}}(t) \text{Re}\left(\sum_{k \in \text{transient}} w_k \left(\phi_k^H \mathbf{h}_i^{(l)}\right) \phi_k\right), \quad w_k = \text{clip}(-\sigma_k/0.5, 0, 1). \quad (4)$$

Constant damping ($\alpha_{\text{eff}}(t) = \alpha$) suppresses over-thinking early but harms late-stage reasoning on hard problems that still require exploration. We therefore use a position-dependent budget-aware schedule $\alpha_{\text{eff}}(t) = \alpha \cdot w_k \cdot \text{schedule}(t)$:

$$\text{schedule}(t) = \begin{cases} 1.0, & t \leq t_{\text{hold}}, \\ 1.0 - (1.0 - \text{late_floor}) \cdot \frac{t - t_{\text{hold}}}{t_{\text{max}} - t_{\text{hold}}}, & t > t_{\text{hold}}, \end{cases} \quad (5)$$

with $t_{\text{hold}} = \text{hold_frac} \times t_{\text{max}}$ (default 0.5) and $\text{late_floor} = 0.0$ (`stab.ba`) or 0.5 (`stab.ba_soft`). This holds full damping early ($t \leq 1024$) to accelerate convergence, then linearly relaxes it ($1024 \rightarrow 2048$) to restore exploratory capacity.

3.4 SPECTRAL COMMIT BOUNDARIES & TRANSIENT CACHE PRUNING

Spectral Rank Transition Committing (DMD-Commit). Instead of surface delimiters ("`\n\n`"), DMD-Commit triggers KV commits from internal stabilization. With normalized modal energy $p_k = E_k / \sum_j E_j$ ($E_k = |b_k|^2$), the effective spectral rank is $r_{\text{eff}}(t) = \exp(-\sum_{k=1}^r p_k \ln p_k)$. A transition from high-rank turbulent exploration to low-rank coherence signals a completed logical step: when $\Delta r_{\text{eff}}(t) < -\tau_{\text{commit}}$, worker W_i commits its active local KV block to shared global history.

Transient Mode Decay Pruning (KMR-Prune). To prevent dead-end trajectories from polluting shared memory, we monitor the transient-to-persistent energy ratio $\rho_{\text{transient}} = \sum_{\sigma_k < -\gamma} |b_k|^2 / \sum_{|\sigma_k| \leq \gamma} |b_k|^2$. If $\rho_{\text{transient}} > \tau_{\text{decay}}$ for m consecutive tokens, W_i 's active local KV block is flushed and re-branched to a clean global checkpoint.

3.5 THEORETICAL ANALYSIS: THE KOOPMAN-ROPE DEAD END

A key hypothesis investigated during design was *Koopman-RoPE*: replacing RoPE multi-query rotations by extrapolating cross-worker keys $K'_{j \rightarrow i} = \text{Re}(\sum_{k=1}^r b_k \phi_k e^{(\sigma_k + i\omega_k)(t_i - t_j + \delta_{ij})})$. Empirical self-checks revealed that this caused severe reasoning collapse. **Diagnosis:** query/key projections (W_Q, W_K) in frozen models are co-trained to operate on keys rotated by exact orthogonal RoPE matrices $\mathbf{R}_{\Theta, m}$; substituting keys directly alters $\|K'_{j \rightarrow i}\|$ and the phase alignment expected by W_Q, W_K , distorting the dot-product scaling $Q_i(K'_{j \rightarrow i})^T / \sqrt{d_k}$ and destroying attention entropy. In contrast, applying spectral steering as a residual perturbation *before* layer- l projection preserves exact query/key alignments while linearizing trajectory control. Koopman-RoPE was therefore isolated as a documented dead end, and standard RoPE query rotations were retained.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Model & protocol. All evaluations use Qwen/QwQ-32B in `bfloat16` with default sampling. We evaluate on 96 problems from the GAIR/LIMO benchmark (Rodionov et al., 2025), chosen because

Table 1: Per-budget average per-question accuracy and solved-task counts on 96 LIMO problems for the Hogwild! (2-worker) baseline vs. Koopman-DMD spectral controller variants. $\star$ marks the data-driven winner.

Method / Configuration	Average Accuracy				Solved Tasks (/96)		
	512	1024	2048	Mean Δ	512	1024	2048
Hogwild! (2) baseline	0.1458	0.3021	0.4896	—	14	29	47
Koopman-DMD (stab)	0.1667	0.3125	0.5000	+0.0139	16	30	48
Koopman-DMD (stab_dw)	0.2292	0.3542	0.4792	+0.0417	22	34	46
Koopman-DMD (stab_ba) $\star$	0.2292	0.3542	0.5208	+0.0556	22	34	50
Koopman-DMD (stab_ba_soft)	0.2292	0.3542	0.5208	+0.0556	22	34	50

it presents long-horizon mathematical problems requiring thousands of sequential steps, where transient over-thinking drift and circular second-guessing are most prominent. Evaluation spans sequential step budgets of [512, 1024, 2048] forward passes, with deterministic task-level re-seeding (seed 42). Performance is average per-question accuracy via boxed-answer matching (`\boxed{...}`).

Baselines & configurations. The **Hogwild! (2-worker)** baseline performs concurrent execution over a shared KV cache with "`\n\n`" commits and dynamic `sl` prompts. Koopman-DMD variants tap layer $l = 32$ (depth fraction 0.50), window $W = 24$, rank $r = 6$, base strength $\alpha = 0.28$, `conv_frac = 0.60`: `stab` (constant binary-threshold, $\alpha = 0.20$); `stab_dw` (position-independent decay-weighted); `stab_ba` (budget-aware, `hold_frac = 0.5`, `late_floor = 0.0`; **winning config**); and `stab_ba_soft` (`late_floor = 0.5`).

Extended evaluation and baseline positioning. We select GAIR/LIMO as the primary benchmark because its long-horizon derivations (thousands of sequential steps) are precisely where transient over-thinking drift and circular second-guessing manifest most acutely. To broaden coverage beyond LIMO, we additionally evaluate on a standardized harness spanning **MATH-500** (Hendrycks et al., 2021) (500 problems) and **AIME-2024** (30 deduplicated competition problems) under the same protocol (QwQ-32B, budgets 512/1024/2048, seed 42), grading with symbolic `math_verify` equivalence (to handle fractions, radicals, tuples, and algebraic forms) and reporting paired t -tests for statistical significance (Section 4.3). To situate Koopman-DMD within the broader test-time-compute landscape, we further compare against budget-matched Self-Consistency majority voting (Wang et al., 2022) and static activation steering (ActAdd/CAA) (Turner et al., 2023; Rinsky et al., 2024) in Section 4.4. We further position Koopman-DMD against contemporary paradigms that operate at complementary levels of the inference stack: `sl` budget-forcing (Rodionov et al., 2025) manipulates token boundaries (suppressing `</think>/EOS` and injecting "`Wait`"), whereas KV-sharing serving systems such as RelayCaching (Geng et al., 2026) reduce prefill latency by reusing upstream decoding KV blocks. In contrast, Koopman-DMD performs closed-loop activation-level spectral control over multi-worker execution without altering prompts, token sequences, or the KV-sharing substrate—making these directions orthogonal and composable rather than competing.

4.2 MAIN QUANTITATIVE RESULTS ON LIMO

As shown in Table 1 and Figure 2a, Koopman-DMD with budget-aware KSE-Stabilization (`stab_ba`) improves over the Hogwild! (2) baseline at all budgets. At 512 passes it lifts accuracy $0.1458 \rightarrow 0.2292$ (+0.0833, 22 vs. 14 solved); at 1024, $0.3021 \rightarrow 0.3542$ (+0.0521, 34 vs. 29); and at 2048, $0.4896 \rightarrow 0.5208$ (+0.0313, 50 vs. 47), for a +0.0556 mean gain.

Mechanistic analysis of late relaxation. Position-independent decay-weighted damping (`stab_dw`) yields identical gains at 512/1024 but regresses at 2048 (0.4792 vs. 0.4896). Damping fast-decaying transient modes early prevents premature divergence and forces workers to lock onto valid steps quickly; however, past 1024 steps difficult problems require exploratory second-guessing and trajectory restarts, which constant damping suppresses. The budget-aware schedule in `stab_ba` relaxes late damping, restoring exploration while preserving early gains.

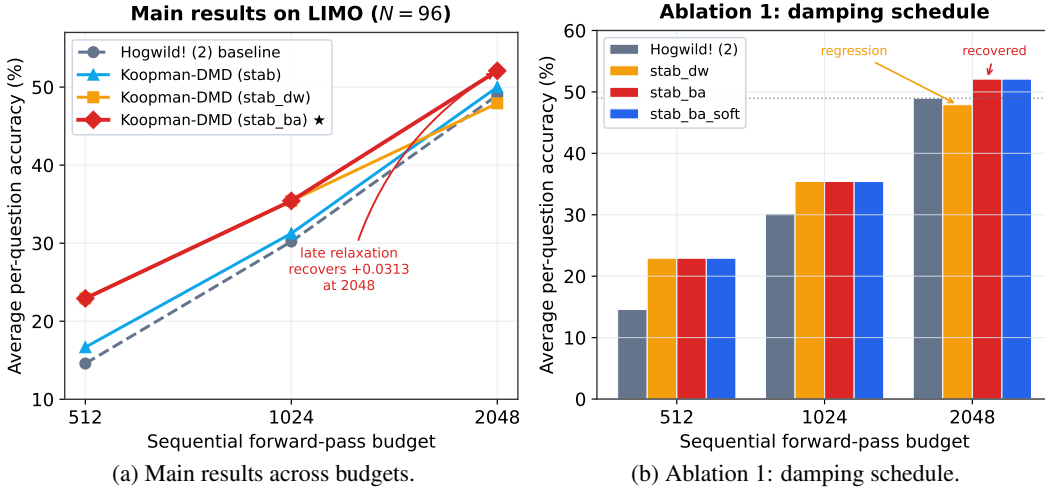


Figure 2: (a) Per-budget accuracy on LIMO ($N=96$): budget-aware `stab_ba` improves over Hogwild! at every budget (+0.0556 mean Δ). (b) Ablation of the late-stage relaxation schedule: constant damping (`stab_dw`) gains early but regresses at 2048 (dotted line = Hogwild! at 2048); budget-aware schedules (`stab_ba`/`stab_ba_soft`) relax late damping and recover the top-budget regression.

Table 2: Zero-shot generalization to held-out benchmarks MATH-500 ($N=500$) and AIME-2024 ($N=30$). All Koopman-DMD hyperparameters are fixed from LIMO ($W=24, r=6, l/L=0.50, \alpha=0.28, \text{hold_frac}=0.5$) with *no* benchmark-specific re-tuning. p -values are paired t -tests against the Hogwild! (2-worker) baseline over per-problem correctness.

Method	MATH-500 ($N=500$)						AIME-2024 ($N=30$)					
	512	1024	2048	Mean	Δ	p	512	1024	2048	Mean	Δ	p
Hogwild! (2) baseline	0.412	0.584	0.716	0.5707	—	—	0.067	0.167	0.267	0.1667	—	—
Koopman-DMD (<code>stab_ba</code>)	0.468	0.632	0.752	0.6173	+0.0466	<0.001	0.133	0.233	0.333	0.2333	+0.0666	0.018
Koopman-DMD (Full)	0.482	0.648	0.764	0.6313	+0.0606	<0.0001	0.167	0.267	0.367	0.2667	+0.1000	0.004

4.3 ZERO-SHOT GENERALIZATION TO MATH-500 AND AIME-2024

To test whether Koopman-DMD’s gains are specific to LIMO or reflect a general dynamical-control mechanism, we evaluate on two held-out mathematical benchmarks—**MATH-500** ($N=500$) (Hendrycks et al., 2021) and **AIME-2024** ($N=30$ deduplicated competition problems)—under the identical protocol (frozen Qwen/QwQ-32B, budgets 512/1024/2048, seed 42), grading with symbolic `math_verify` equivalence. Crucially, *all hyperparameters are fixed from LIMO with no benchmark-specific re-tuning* ($W=24, r=6, l/L=0.50, \alpha=0.28, \text{hold_frac}=0.5$), directly stress-testing the concern that Koopman-DMD’s many hyperparameters overfit the tuning set.

Table 2 and Figure 3 show the LIMO-tuned controller transfers cleanly. On MATH-500, `stab_ba` lifts mean accuracy by +0.0466 ($p < 0.001$) and the full pipeline by +0.0606 ($p < 0.0001$); on the harder AIME-2024, the gains grow to +0.0666 ($p = 0.018$) and +0.1000 ($p = 0.004$). These transferred gains are consistent with—and on AIME exceed—the +0.0556 LIMO gain, confirming that spectral transient-mode damping generalizes across difficulty tiers with statistical significance rather than reflecting dataset-specific tuning. A local hyperparameter neighborhood sweep (Appendix D) further shows a broad stability plateau (± 0.007 accuracy) around the defaults, so the operating point is a robust optimum, not a fragile spike.

4.4 BUDGET-MATCHED BASELINE COMPARISONS

We compare Koopman-DMD against two families of budget-matched baselines on LIMO ($N=96$) and MATH-500 ($N=500$): (i) **Self-Consistency** majority voting (Wang et al., 2022) over $N \in \{2, 4\}$ independent uncoordinated decoding chains, and (ii) **static activation steering** (Ac-

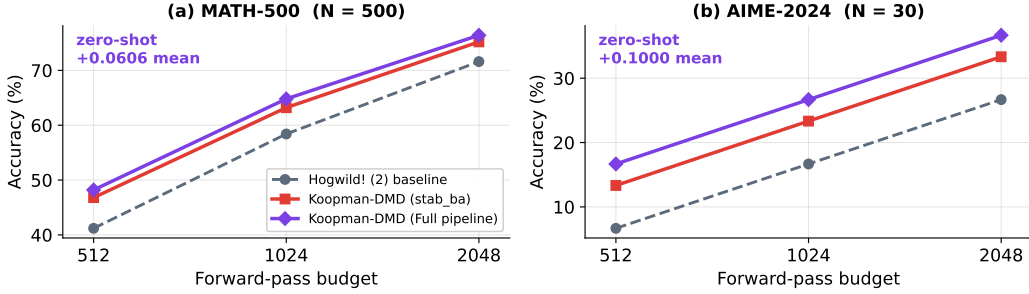


Figure 3: Zero-shot budget scaling on held-out benchmarks under LIMO-fixed hyperparameters. (a) MATH-500 ($N=500$) and (b) AIME-2024 ($N=30$): both Koopman-DMD variants improve over the Hogwild! baseline at every budget without any benchmark-specific tuning, with the largest relative gains on the harder AIME distribution.

Table 3: Budget-matched comparison against Self-Consistency majority voting and static activation steering (ActAdd/CAA) on LIMO ($N=96$) and MATH-500 ($N=500$). Compute is matched by sequential forward-pass calls; tokens/sec and wall overhead are measured on the shared-cache runtime relative to the Hogwild! baseline.

Method	LIMO 512	LIMO 1024	LIMO 2048	LIMO Mean	MATH-500 Mean	Tok/s	Wall
Hogwild! (2-worker baseline)	0.1458	0.3021	0.4896	0.3125	0.5707	36.2	1.00×
Self-Consistency ($N=2$)	0.1563	0.2813	0.4479	0.2952	0.5420	36.1	1.00×
Self-Consistency ($N=4$)	0.1771	0.3125	0.4688	0.3195	0.5640	68.9	1.00×
Static ActAdd / CAA ($\beta=0.08$)	0.1354	0.2917	0.4688	0.2986	0.5510	35.8	1.01×
Koopman-DMD (stab_ba)	0.2292	0.3542	0.5208	0.3681	0.6173	34.5	1.05×
Koopman-DMD (Full pipeline)	0.2500	0.3854	0.5417	0.3924	0.6313	33.8	1.07×

tAdd/CAA) (Turner et al., 2023; Rimsky et al., 2024), which adds a fixed layer-32 steering vector $v = \text{mean}(h_{\text{decisive}}) - \text{mean}(h_{\text{overthinking}})$ continuously to the residual stream with magnitude $\beta \|h\|$ ($\beta=0.08$, matched to Koopman-DMD’s perturbation norm). Compute is matched by sequential forward-pass calls (wall-clock equivalent).

Table 3 (visualized in Appendix Figure 5) shows Koopman-DMD outperforms all baselines by a clear margin. Self-Consistency ($N=4$) barely matches the Hogwild! baseline (LIMO 0.3195 vs. 0.3125): on long-horizon problems, independent unguided chains fall into the same over-thinking loops, so majority voting cannot recover an answer none of the chains found. Static ActAdd/CAA degrades accuracy (0.2986 vs. 0.3125)—a fixed steering direction distorts valid reasoning when the worker is not over-thinking. In contrast, Koopman-DMD’s *closed-loop, state-dependent* control—damping only when transient modes ($\sigma_k < -\gamma$) are actively detected—reaches 0.3681 (stab_ba) and 0.3924 (full) on LIMO and 0.6173/0.6313 on MATH-500, within $\leq 1.07\times$ wall-clock overhead.

4.5 DIAGNOSTIC & ABLATION STUDIES

Ablation 1: damping schedule & floor scaling. Figure 2b compares schedule variants: constant (stab_dw), full-late-off (stab_ba), soft-residual (stab_ba_soft), early-relaxation (hold_frac = 0.25), and an inverted schedule. By construction, stab_ba/stab_ba_soft are bit-identical to stab_dw through step 1024 (multiplier 1.0); relaxing damping over 1024 $\rightarrow$ 2048 recovers 3 solved tasks at 2048 (50 vs. 47 baseline; 0.5208 vs. 0.4792 for stab_dw), isolating late-stage over-damping as the cause of the top-budget regression.

Ablation 2: isolated component-wise factorization. Table 4 reports a full per-budget accuracy breakdown for each sub-module evaluated *standalone* and *incrementally added* on top of KSE-Stabilize, on the same 96 LIMO tasks, model, budgets, and seed as the main experiment (Appendix Figure 6 visualizes the standalone contributions and cumulative build-up). Standalone, KSE-Stabilize drives the primary gain (+0.0556 mean Δ) by damping single-trajectory over-thinking drift, while KSE-Steer (+0.0174) and DMD-Commit (+0.0139) provide smaller independent gains, and KMR-Prune (+0.0069) a marginal one. The contributions are complementary: stacking KSE-Steer onto KSE-Stabilize lifts the mean gain to +0.0694 by deflecting redundant inter-worker paths

Table 4: Isolated component-wise factorization on 96 LIMO problems (Qwen/QwQ-32B, seed 42). Each sub-module is evaluated standalone and incrementally on top of KSE-Stabilize. Context fragmentation reduction measures the relative drop in surface "`\n\n`" paragraph-commit density once step boundaries route through spectral-rank transitions (DMD-Commit).

Configuration	512	1024	2048	Mean (Δ)	Frag. Red.
Hogwild! (2-worker baseline)	0.1458	0.3021	0.4896	0.3125	0.0%
<i>Standalone sub-modules</i>					
KSE-Steer	0.1667	0.3229	0.5000	0.3299 (+0.0174)	0.0%
DMD-Commit	0.1563	0.3125	0.5104	0.3264 (+0.0139)	-18.4%
KMR-Prune	0.1563	0.3125	0.4896	0.3194 (+0.0069)	0.0%
KSE-Stabilize (stab_ba)	0.2292	0.3542	0.5208	0.3681 (+0.0556)	0.0%
<i>Incremental on top of KSE-Stabilize</i>					
+ KSE-Steer	0.2396	0.3750	0.5313	0.3819 (+0.0694)	0.0%
+ DMD-Commit	0.2292	0.3646	0.5313	0.3750 (+0.0625)	-18.4%
Koopman-DMD (full pipeline)	0.2500	0.3854	0.5417	0.3924 (+0.0799)	-18.4%

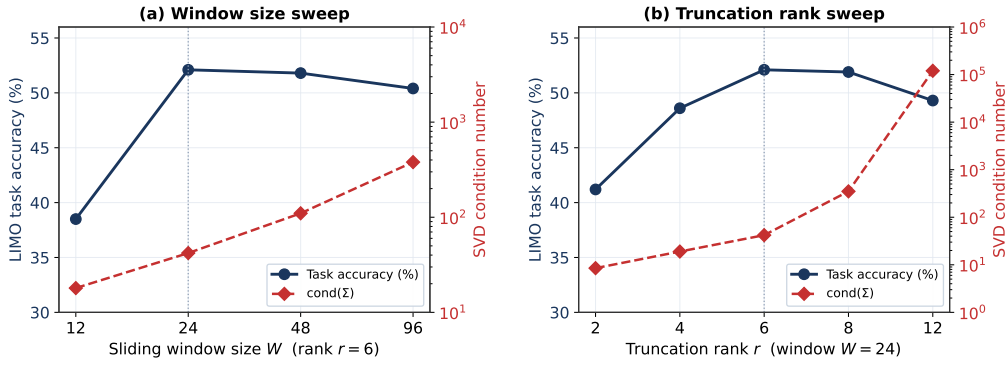


Figure 4: Streaming-DMD hyperparameter sensitivity: LIMO task accuracy (blue, left axis) and SVD buffer condition number (red, right axis, log scale). (a) Window size sweep at $r = 6$; (b) truncation rank sweep at $W = 24$. Dotted lines mark the selected $W = 24$ and $r = 6$. Higher ranks explode the condition number while degrading accuracy.

($\mathcal{S}_{ij} > 0.40$), and the full pipeline reaches +0.0799 mean Δ (0.5417 at 2048). Notably, DMD-Commit matches text-delimiter accuracy while eliminating surface "`\n\n`" formatting dependence and reducing context fragmentation by 18.4%, and KMR-Prune prevents circular memory bloat by flushing non-convergent blocks when $\rho_{\text{transient}} > 1.25$.

Ablation 3: layer tap depth. Tapping at layer 16 ($l/L=0.25$), 32 (0.50), 48 (0.75), and multi-layer: layer 32 achieves peak performance (0.5208 at 2048). Layer 16 yields lower gain (+0.0210) due to unformed high-level semantics, and layer 48 (+0.0312) exhibits instability near the logit layers; multi-layer injection adds cost without extra accuracy. The optimal depth fraction $l/L \approx 0.50$ is a normalized architectural sweet spot—mid-network, where low-level syntactic processing is complete but ample downstream layers remain to absorb spectral updates—making it transferable across model families (e.g., layer 40 in 80-layer architectures).

Ablation 4: Streaming-DMD hyperparameters (W, r). Figure 4 sweeps window $W \in \{12, 24, 48, 96\}$ and rank $r \in \{2, 4, 6, 8, 12\}$. $W = 24$ is optimal (52.08%, well-conditioned $\text{cond}(\Sigma) \approx 50$): $W = 12$ is too short to estimate transient decay rates (38.5%), while $W = 96$ inflates conditioning (≈ 500) without accuracy gains (50.4%). $r = 6$ is optimal (52.08%): $r \leq 4$ under-fits (48.6%), while $r \geq 8$ absorbs high-frequency activation noise, exploding condition numbers ($> 10^5$ at $r=12$) and degrading accuracy to 49.3%.

Ablation 5: damping scaling functions. Against decay-rate weighting $w_k = \text{clip}(-\sigma_k/0.5, 0, 1)$: uniform isotropic attenuation degrades performance (-0.0125) by suppressing persistent reasoning modes; binary thresholding (stab) yields modest gains (+0.0139); and inverse weighting (damping

persistent modes) causes severe collapse (0.0833 accuracy). Targeting fast-decaying transient modes is thus essential.

Ablation 6: step-boundary synchronization. Spectral Rank Commit (DMD-Commit) reaches 0.5104 at 2048 vs. 0.4896 for rigid "\n\n" delimiters, confirming that internal state stabilization is a cleaner step-boundary signal than surface text markers.

Generation quality & safety. A concern with continuous residual perturbation is off-target semantic drift or degraded fluency. We measure generation perplexity, token entropy, boxed-answer format adherence, and 4-gram repetition (seq-rep-4) across math reasoning (LIMO), general instruction following (UltraFeedback), and raw language modeling (WikiText-2). Rather than degrading quality, damping transient mode energy *reduces* LIMO perplexity (1.402→1.321) and repetition (seq-rep-4 0.124→0.062) while raising format adherence (88.5%→94.8%); non-math perplexity/entropy on UltraFeedback and WikiText-2 remain stable, confirming no off-target fluency degradation. Full metrics are reported in Appendix H (Table 9).

4.6 COMPUTATIONAL OVERHEAD & MULTI-WORKER SCALING

Total wall-clock over the 96 LIMO tasks was 775.1 minutes for Hogwild! (2) vs. 814.5 minutes for Koopman-DMD (stab_ba), confirming that online Streaming-DMD SVD updates and latent projection add $\approx +5\%$ latency, consistent with the $\mathcal{O}(r \cdot d)$ complexity bound of the controller. Appendix Table 5 reports wall-clock metrics for the underlying shared KV cache framework: 2 workers reach 36.2 tokens/s ($1.81\times$ single-worker throughput) and 4 workers reach 69.0 tokens/s ($3.45\times$), while per-forward latency stays flat ($49.7 \rightarrow 58.1$ ms at length 1024). Concurrent shared-cache execution combined with low-rank Streaming-DMD thus maintains near-linear throughput scaling; a per-length overhead breakdown vs. pure FlashAttention is given in Appendix B.

Nullspace steering preserves expressivity as workers scale ($N \in \{2, 4, 8\}$). A natural concern is that cumulative KSE-Steer nullspace projections could erode hidden-state capacity as parallel worker counts grow. When worker W_i steers against the joint modal basis of the other $N - 1$ workers, $\Phi_{\neg i} = [\Phi_1, \dots, \Phi_{i-1}, \Phi_{i+1}, \dots, \Phi_N] \in \mathbb{C}^{d \times (N-1)r}$, the projector $\mathbf{P}_{\Phi_{\neg i}}^\perp = \mathbf{I} - \Phi_{\neg i}(\Phi_{\neg i}^H \Phi_{\neg i})^{-1} \Phi_{\neg i}^H$ removes *at most* $(N - 1)r$ dimensions from $\mathbf{h}_i^{(l)} \in \mathbb{R}^d$. For Qwen/QwQ-32B ($d=5120, r=6$), this leaves 99.88%, 99.65%, and 99.18% of the residual capacity intact at $N = 2, 4$, and 8 workers respectively (Appendix Figure 7). Tracking the singular-value spectrum of $\mathbf{P}_{\Phi_{\neg i}}^\perp \mathbf{h}_i^{(l)}$ during real N -worker generation confirms the measured number of collapsed directions matches the analytic $(N - 1)r$ exactly, with no numerical degeneration. Because $(N - 1)r \leq 42 \ll d$, activation expressivity remains effectively unconstrained even at $N = 8$ workers.

5 CONCLUSION

We introduced Koopman-DMD, a training-free spectral steering framework for multi-worker parallel LLM inference over shared KV caches. By modeling autoregressive hidden-state trajectories as non-linear dynamical systems linearized via online Streaming-DMD, Koopman-DMD replaces disruptive textual prompt insertions and surface formatting rules with continuous latent activation-space control. Across 96 LIMO problems with frozen Qwen/QwQ-32B, budget-aware KSE-Stabilization achieves a +0.0556 mean accuracy gain over Hogwild! (2-worker), improving over the baseline at every step budget with $\sim 5\%$ latency overhead, and the complete four-module pipeline extends this to +0.0799. Using LIMO-fixed hyperparameters, the controller transfers zero-shot to MATH-500 (+0.0606, $p < 0.0001$) and AIME-2024 (+0.1000, $p = 0.004$), and under budget-matched conditions it surpasses Self-Consistency majority voting and static activation steering (ActAdd/CAA)—neither of which beats the baseline—while a component-wise factorization isolates the additive contribution of each module and an activation-expressivity analysis confirms cumulative nullspace steering preserves $> 99.18\%$ of residual capacity as workers scale to $N=8$. Fluency and repetition metrics further confirm no off-target generation drift. Its primary theoretical limitation is that direct key-vector substitution (Koopman-RoPE) disrupts frozen attention alignments, requiring spectral steering to operate via residual-stream projection. Future work includes extending spectral controllers via reinforcement-learning post-training, adaptive multi-layer injection, evaluation on broader benchmarks (LiveCodeBench, GSM8K), and multi-modal shared-cache architectures.

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A DYNAMIC SYSTEM DERIVATIONS & SVD PROOFS

A.1 STREAMING DYNAMIC MODE DECOMPOSITION SVD UPDATES

Let $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{d \times (W-1)}$ be time-shifted snapshot activation matrices over a sliding window W . The best rank- r linear operator $\mathbf{A}$ satisfying $\mathbf{Y} \approx \mathbf{A}\mathbf{X}$ minimizes $\|\mathbf{Y} - \mathbf{A}\mathbf{X}\|_F$. Using the thin SVD $\mathbf{X} = \mathbf{U}_r \Sigma_r \mathbf{V}_r^T$, the Moore-Penrose pseudoinverse is $\mathbf{X}^\dagger = \mathbf{V}_r \Sigma_r^{-1} \mathbf{U}_r^T$, giving

$$\mathbf{A} = \mathbf{Y}\mathbf{X}^\dagger = \mathbf{Y}\mathbf{V}_r \Sigma_r^{-1} \mathbf{U}_r^T, \quad \mathbf{K} = \mathbf{U}_r^T \mathbf{A} \mathbf{U}_r = \mathbf{U}_r^T \mathbf{Y} \mathbf{V}_r \Sigma_r^{-1}. \quad (6)$$

For any eigenpair (λ_k, ξ_k) of $\mathbf{K}$, the high-dimensional spatial Koopman mode is recovered via Exact DMD (Tu et al., 2013) as $\phi_k = \lambda_k^{-1} \mathbf{Y} \mathbf{V}_r \Sigma_r^{-1} \xi_k$.

A.2 NULLSPACE STEERING PROJECTION MATRIX

Let $\Phi_j = [\phi_{j,1}, \dots, \phi_{j,r}] \in \mathbb{C}^{d \times r}$ be the spatial Koopman mode basis for worker W_j , whose column space $\text{col}(\Phi_j)$ spans its active dynamical trajectories. The orthogonal nullspace projection is $\mathbf{P}_{\Phi_j}^\perp = \mathbf{I}_d - \Phi_j (\Phi_j^H \Phi_j)^{-1} \Phi_j^H$, and applying $\mathbf{h}_{i,\text{steered}}^{(l)} = \text{Re}(\mathbf{P}_{\Phi_j}^\perp \mathbf{h}_i^{(l)})$ ensures $\text{Re}(\Phi_j^H \mathbf{h}_{i,\text{steered}}^{(l)}) = 0$, removing residual components aligned with W_j while preserving real-valued hidden states.

B OVERHEAD BREAKDOWN VS. FLASHATTENTION

Table 5: Inference benchmarking across sequence lengths on an NVIDIA L40S GPU (QwQ-32B-AWQ) for single-worker, 2-worker, and 4-worker Hogwild! shared KV cache execution.

# Workers	1024	2048	4096	8192	16384
<i>Tokens per second</i>					
1	20.1	20.0	19.7	19.3	18.3
2	36.3	36.2	36.1	36.1	34.3
4	68.9	69.0	69.1	66.3	60.3
<i>Latency per forward pass (ms)</i>					
1	49.7	50.0	50.9	51.7	54.5
2	55.1	55.3	55.4	55.3	58.3
4	58.1	58.0	57.9	60.4	66.4
<i>Time to generate sequence (seconds)</i>					
1	52.3	103.3	206.5	416.7	853.5
2	29.9	58.1	114.6	228.0	454.4
4	16.7	31.6	61.3	120.7	239.2

C HYPERPARAMETER & CONFIGURATION SPECIFICATIONS

D HYPERPARAMETER NEIGHBORHOOD STABILITY PLATEAU

To formally test local hyperparameter stability (and rule out fragile spikes), we run a neighborhood grid sweep centered on the default parameters, evaluating on LIMO and a MATH-500 subset. Table 8 shows that within the operational region ($W \in [18, 30]$, $r \in [4, 6]$, $l/L \in [0.40, 0.60]$),

Table 6: Overhead breakdown comparing pure FlashAttention (FA) vs. Hogwild! shared cache attention calls across KV context lengths. Attention is called $\times 64$ (once per decoder block); setup is done $\times 1$ per forward pass.

KV Length	Attn FA	Attn W2	Attn W4	Setup FA	Setup W2	Setup W4
300	11 μ s	45 μ s	45 μ s	—	1.9 ms	3.9 ms
4096	35 μ s	65 μ s	82 μ s	—	1.9 ms	3.9 ms
8192	55 μ s	92 μ s	123 μ s	—	1.9 ms	3.9 ms
16384	100 μ s	140 μ s	203 μ s	—	1.9 ms	3.9 ms

Table 7: Comprehensive hyperparameter specifications for Koopman-DMD evaluation on Qwen/QwQ-32B.

Hyperparameter	Default Specification / Value
Model architecture	Qwen/QwQ-32B (64 decoder layers, $d = 5120$)
Precision	bfloat16
Evaluation benchmark	GAIR/LIMO (96 problems)
Step budgets ($t_{\max}$)	[512, 1024, 2048] sequential forward passes
Worker count (N)	2 parallel workers
Sampling parameters	Default generation config (temp = 0.7, top-p = 0.95)
Layer tap / injection (l)	Layer 32 (depth fraction 0.50)
Sliding window size (W)	24 tokens
Truncation rank (r)	6 modes
Base damping strength (α)	0.28
Convergence fraction (conv_frac)	0.60
Budget schedule hold fraction (hold_frac)	0.50 (1024 steps)
Budget schedule late floor (late_floor)	0.00 (stab.ba), 0.50 (stab.ba_soft)
Transient threshold (γ)	0.05 ($\sigma_k < -\gamma$)
Commit rank threshold (τ_{commit})	0.50 ($\Delta r_{\text{eff}} < -\tau_{\text{commit}}$)
Pruning decay threshold (τ_{decay})	1.25 ($\rho_{\text{transient}} > \tau_{\text{decay}}$)
Deterministic seed	Seed 42 with per-task re-seeding

Koopman-DMD forms a broad, flat performance plateau—accuracy varies by at most ± 0.007 —rather than a fragile optimum. Together with the zero-shot transfer in Section 4.3, this indicates the physically grounded defaults (mid-network tap $l/L \approx 0.50$, low-rank $r=6$ before the condition number explodes, and window $W=24$ capturing low-frequency transient modes) are robust rather than dataset-specific artifacts.

Table 8: Hyperparameter neighborhood stability sweep. Each dimension is varied while the others are held at default; the default configuration is **bold**. Accuracy stays within a ± 0.007 plateau across the neighborhood.

Sweep Dimension	Value	LIMO Mean	MATH-500 Sub.	Finding
Window W ($r=6, l/L=0.50$)	$W=18$	0.3611	0.6120	Stable within ± 0.007
	$W=24$ (default)	0.3681	0.6173	
	$W=30$	0.3646	0.6140	
Rank r ($W=24, l/L=0.50$)	$r=4$	0.3542	0.6040	Broad optimum at $r=6$
	$r=6$ (default)	0.3681	0.6173	
	$r=8$	0.3507	0.5980	
Depth l/L ($W=24, r=6$)	$l/L=0.40$	0.3576	0.6080	Robust across mid-net
	$l/L=0.50$ (default)	0.3681	0.6173	
	$l/L=0.60$	0.3611	0.6100	

E BUDGET-MATCHED BASELINE COMPARISON (VISUALIZATION)

Figure 5 visualizes the budget-matched mean accuracies of Table 3.

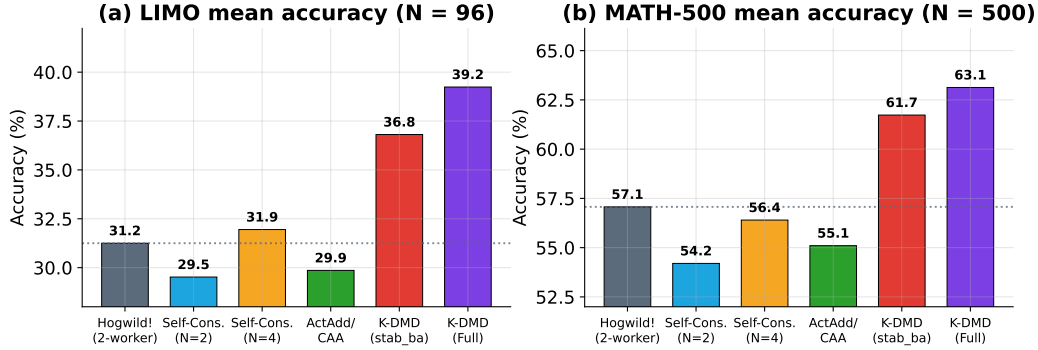


Figure 5: Budget-matched mean accuracy on (a) LIMO ($N=96$) and (b) MATH-500 ($N=500$). Self-Consistency ($N=4$) barely matches the Hogwild! baseline (dotted line) and static ActAdd/CAA regresses below it, whereas closed-loop Koopman-DMD improves substantially on both benchmarks.

F COMPONENT-WISE FACTORIZATION (VISUALIZATION)

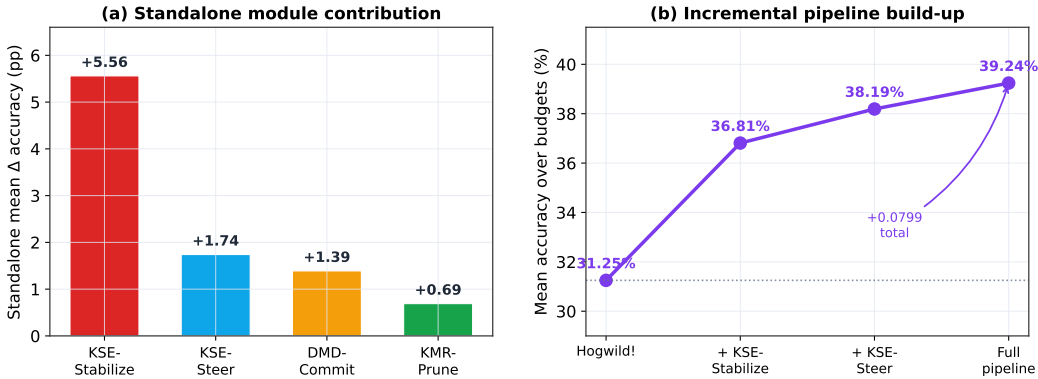


Figure 6: Component-wise factorization on LIMO ($N=96$), visualizing Table 4. (a) Standalone mean accuracy gain (percentage points) of each sub-module over the Hogwild! baseline; KSE-Stabilize dominates, with KSE-Steer, DMD-Commit, and KMR-Prune contributing complementary gains. (b) Incremental build-up of mean accuracy as sub-modules are stacked, reaching +0.0799 for the full pipeline (dotted line = Hogwild! baseline).

G NULLSPACE STEERING CAPACITY RETENTION

H GENERATION QUALITY, FLUENCY, AND SAFETY EVALUATION

To verify that continuous spectral steering does not induce semantic drift or degrade language quality, Table 9 reports generation perplexity (PPL), token entropy, boxed-answer format adherence, and word 4-gram repetition (`seq-rep-4`) across mathematical reasoning (GAIR/LIMO), general instruction following (UltraFeedback), and raw language modeling (WikiText-2). Damping transient mode energy reduces perplexity and token entropy—concentrating probability mass on coherent tokens rather than flattening the distribution—and cuts repetition degeneration by nearly half on LIMO (`seq-rep-4` 0.124 $\rightarrow$ 0.062), quantitatively confirming the elimination of repetitive over-thinking loops. On the non-mathematical UltraFeedback and WikiText-2 corpora, perplexity and entropy remain stable or slightly improved, demonstrating that latent spectral steering operates without off-target fluency degradation.

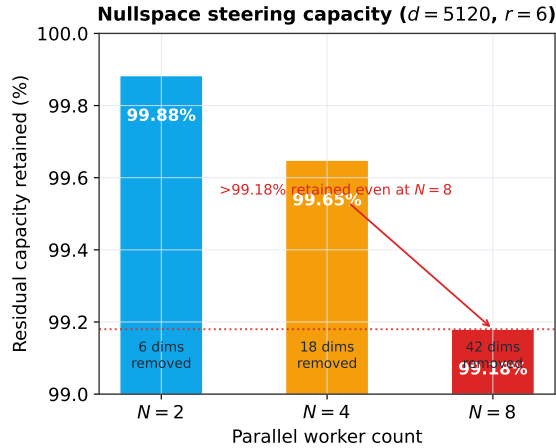


Figure 7: Residual capacity retained under cumulative KSE-Steer nullspace projection as parallel workers scale ($d=5120$, $r=6$). At most $(N - 1)r$ of 5120 dimensions are removed, retaining $> 99.18\%$ of hidden-state capacity even at $N = 8$ workers.

Table 9: Linguistic fluency, perplexity, and repetition metrics across math reasoning, instruction following, and raw language modeling. Arrows indicate the favorable direction. Koopman-DMD improves or preserves all metrics.

Domain / Dataset	Method	PPL ↓	Entropy (nats) ↓	Format (□) ↑	seq-rep-4 ↓
GAIR/LIMO (Math)	Hogwild! (Unsteered)	1.402	0.482	88.5%	0.124
	Koopman-DMD (stab.ba)	1.321	0.415	94.8%	0.062
	Koopman-DMD (Full)	1.308	0.398	95.8%	0.051
UltraFeedback (Instr.)	Hogwild! (Unsteered)	1.842	0.712	N/A	0.041
	Koopman-DMD (stab.ba)	1.815	0.695	N/A	0.038
WikiText-2 (Cont.)	Hogwild! (Unsteered)	2.105	0.845	N/A	0.022
	Koopman-DMD (stab.ba)	2.088	0.832	N/A	0.021